

PRIME MINISTER OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
BUSEGA DISTRICT COUNCIL EXAMINATION
FORM ONE TERMINAL EXAMINATION
SUBJECT: PHYSICS MARKING SCHEME

SECTION A 15MARKS

1. Answer 01marks @ = 10marks

GEPAM science hub

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i.	ii.	iii.	iv.	v.	vi.	vii.	viii.	ix.	x.
B	B	B	D	C	C	A	B	E	B

2. Answers 01mark @ = 05marks

LIST A	i.	ii.	iii.	iv.	v.
LIST B	C	F	I	B	E

SECTION B (70 MARKS)

3. (a) The following are the importances of studying physics (choose any five points)

- i. Understanding the nature.
- ii. Development of problem - solving skills. **01mark@ = 05marks**
- iii. Technological development
- iv. Careers opportunities like teachers, doctors , technicians, nurses and so on
- v. Understanding universe.
- vi. Daily life application like cooking and driving
- vii. Understanding energy and resources
- viii. Innovation and creativity

NB: consider other points relevant to the question

(b) 01mark@ = 05marks

MASS	WEIGHT
i. Is a quantity of an object	i. Is the gravitational force that acting on an object
ii. The SI unit of mass is kg	ii. The SI unit is NEWTON
iii. It is measured by beam balance	iii. It is variable.
iv. It is constant.	iv. It is measured by spring balance
v. Its symbol is M	v. Its symbol is W

4. (a) **01mark@ = 05marks**

- i. Don't run, play or fighting in the laboratory.
- ii. Keep your hand dry when using electrical apparatus.
- iii. Never touch equipment, chemical or electrical devices without permission.
- iv. Wear protective equipment when required.
- v. Report any accident, injury or break tools

NB: consider other points relate to the questions.

(b) The following are types of variables

- i. Independent variable **1.6 mark @ = 05marks**

- ii. Dependant variable
iii. Controlled variable
5. (a) Force: is the push or pull experienced by an object. **02marks**
(b) Archimedes' principle states that "Any object partially or totally immersed in fluid experiences an up thrust which is equal to the weight of the fluid displaced by the object" **08 marks**
6. (a) Error: is the deviation or mistake to actual value. Or is the difference between measured value and the true value. **02 marks**
(b) Answers
i. Damage during use (systematic error)
ii. Instrumental error
iii. Poor storage (environmental error) **08 marks**
iv. Human factor (human error)
7. (a) Attraction force: is the force that pulls two objects towards each other.
While
Repulsion force; is the force that pushes object away from each other. **02 marks@**
= 04 marks

(b) answer 06 marks

SCALAR QUANTITY		VECTOR QUANTITY	
i.	It has magnitude only	i.	It has both magnitude and direction
ii.	It does not have direction	ii.	Represented by using arrows
iii.	It describes by a number and units	iii.	It must state how much and which direction.

8. (a) solution **05 marks**
Initial volume = 78cm^3 , Final volume = 105cm^3 , volume used =
?Type equation here.

From

$$\text{Volume used} = \text{final volume} - \text{initial volume}$$

$$\text{Volume used} = 105\text{cm}^3 - 78\text{cm}^3$$

$$\text{Volume Used} = 27\text{cm}^3$$

- (b) Solution; **05 marks**
Length = 23cm, width = 9cm, height = 8cm volume = ?

$$\text{Volume} = \text{length} \times \text{width} \times \text{height}$$

$$\text{Volume} = 23\text{cm} \times 9\text{cm} \times 8\text{cm}$$

$$\text{Volume} = 1656\text{cm}^3 \text{ or } 0.001656\text{m}^3$$

9. (a) Solution; **05 marks**
Weight on the earth (W_e) = 10,000N
Weight in planet X (W_x) = 300N
Acceleration due to gravity on planet X (g_x) = ?
Acceleration due gravity on the earth (g_e) = 10N/K
(W_e) = $M_e (g_e)$
(W_e) = $M_e \times 10\text{N/Kg}$
 $10,000\text{N} = M_e \times 10\text{N/Kg}$

$$\frac{10,000N}{10N/kg} = M_e$$

The mass of the earth = 1000Kg = the mass of the planet X

From;

$$(g_x) = \frac{W_x}{M_x}$$

$$(g_x) = \frac{300N}{1000Kg}$$

$$(g_x) = 0.3 \text{ N/Kg}$$

The gravitational due the planet X is 0.3N/Kg

(b) Answer (five points) 01mark @ = 05marks

- i. It is always attractive.
- ii. It act between all masses
- iii. It follows the universe – square law
- iv. It is a long range force (any five)
- v. It is proportional to masses
- vi. It is a weak force
- vii. It act along the line joining centres

NB: (consider other points related to question)

SECTION C (15 MARKS)

10. (a)

- i. The law of floatation; State that “A floating object float displaces its own weight of fluid in which it floats” **02marks**
- ii. Pascal principal state that “When a pressure a pressure applied at any point on the surface of a fluid contained in closed container the pressure is transmitted undiminished(equally) to all parts of fluid and to the walls of the container” **03 marks**

(b) Answers (five points) (Each student should be able to explain briefly) **02marks @ = 10marks**

- i. Mechanics -it deals with objects which are either stationary or in motion, forces and its impact
- ii. Optics-deals with light and its application
- iii. Heat – form of energy that can be transferred from one point to another by conduction, conduction and radiation.
- iv. Light- form of energy which deals with sensation of vision
- v. Electronics- deals with new technology
- vi. Astronomy-deals with the study of universe
- vii. Electromagnetism – deals with interaction between electric and magnet
- viii. Geophysics -deals with physical process and property of earth and its surrounding

